

Internet-Scale VQA + Robot Action Data



Q: What is happening in the image?

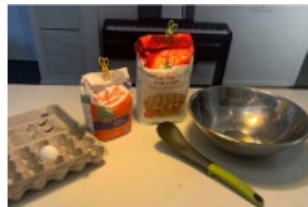
A: 311 423 170 55 244

A grey donkey walks down the street.

Q: Que puis-je faire avec ces objets?

A: 3455 1144 189 25673

Faire cuire un gâteau.



Q: What should the robot do to **<task>**?

A: 132 114 128 5 25 156

$\Delta\text{Translation} = [0.1, -0.2, 0]$

$\Delta\text{Rotation} = [10^\circ, 25^\circ, -7^\circ]$

Vision-Language-Action Models for Robot Control

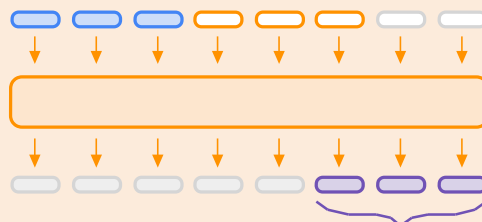
Q: What should the robot do to **<task>**? A: ...



RT-2

ViT

Large Language Model



A: 132 114 128 5 25 156

De-Tokenize

$\Delta^T = [0.1, -0.2, 0]$

$\Delta^R = [10^\circ, 25^\circ, -7^\circ]$

Robot Action

Co-Fine-Tune

Deploy

Closed-Loop Robot Control



Put the strawberry into the correct bowl



Pick the nearly falling bag



Pick object that is different